Suppression of Current Quantization Effects for Precise Current Control of SPMSM Using Dithering Techniques and Kalman Filter

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***Abstract*—High-precision current control is demanded for many mechatronic systems to improve the static and dynamic performances. However, current measurement error inherent in the measurement system degrade the control accuracy, since the motor currents are not guaranteed to follow the desired references. In this paper, quantization error caused by analog-to-digital converters is studied. The dithering techniques combined with Kalman filter are proposed to suppress the quantization effects. First, two dithered systems, including the subtractively dithered and nonsubtractively dithered systems, are designed to whiten the quantization error. In the design of nonsubtractively dithered system, the probability characteristic of the metering noise is utilized to minimize the measurement noise level. Then, Kalman filter is designed to estimate the real current signals from the dithered current measurements. Since the variance of the total measurement error is theoretically calculable according to the proposed dithering techniques, Kalman gain can be determined analytically. Finally, simulations and experiments are performed to verify the effectiveness of the proposed approaches using a high-precision positioning system.**

***Index Terms*—Analog-to-digital converter (ADC), current control, dithering techniques, Kalman filter, motion control, quantization.**

I. Introduction

PRECISE current-controlled permanent magnet synchronous motor (PMSM) drive systems require accurate current measurement to produce high static and dynamic performances [1]–[3]. However, current measurement errors caused by current sensors and analog-to-digital converters (ADCs) limit the current control performance, since the real current signals are not guaranteed to follow the desired references accurately. This would induce torque ripples and deteriorate the control performance especially when a drive system works at low load [4], [5]. Furthermore, current quantization error may also affect the accuracy of rotor position estimation when high-frequency signal injection method is applied for motor sensorless control [6], [7]. Therefore, understanding and suppressing the effects caused by the current measurement errors have attracted a great deal of attention [8], [9].

Fig. 1 shows a typical path of current measurement. Current signals, which are measured by current sensors, are converted into digital signals via ADCs after the amplitude amplification and noise filtering. During the process, quantization error and metering noise containing scaling error, offset error, and thermal noise are introduced into the control system. The quantization error arises between the input and the output of ADCs because the analog signal may assume any value within the input range, while the output data are finite-precision samples [10]. This kind of error behaves as highly colored noise and cannot be simply suppressed by low-pass filter. In contrast, the scaling error and the offset error are usually caused by aging deterioration, and can be corrected by periodic calibration. In this paper, the current measurement system is assumed to be well calibrated and the scaling error and the offset error considered in [5] and [8] are ignored. Fig. 2 shows the block diagram of the measurement process, where η denotes the metering noise and q denotes the quantization error. An example to demonstrate the characteristic of current measurement error is shown in Fig. 3. Here, the original current signal i is a sinusoidal signal with the magnitude and frequency of 1 A and 5 Hz. The quantization step is 0.1 A and the metering noise is uniform in $[-0.025,\ 0.025]$ A. The measurement error and the corresponding power spectral density (PSD) are also shown in the figure. The harmonics caused by the quantization are observable all over the frequency domain. Torque ripples may therefore be generated by these harmonics. Usually, mechanical system can work as a low-pass filter to reduce the high-order harmonics. Low-order harmonics, however, cannot be effectively reduced. In order to suppress current quantization effects, high-resolution ADCs can be used to improve the quality of measured signals. However, not only would the implementation cost be increased, but also the sample rate of ADCs will be limited due to the trade-off between resolution and sample rate [10]. In [1],

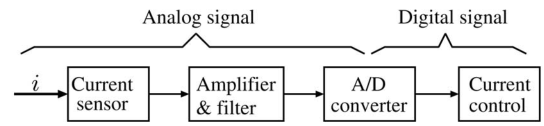

Fig. 1. Path of current measurement.

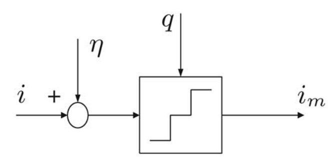

Fig. 2. Block diagram of current measurement.

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Assuming that the input is always within the measurement range of ADCs, the output y can be expressed in terms of the input x as

$$y = Q(x) = \Delta \left\lfloor \frac{x}{\Delta} + \frac{1}{2} \right\rfloor \tag{6}$$

where Q

The subtractively dithered system is ideal in the sense that not only can e_i be rendered independent of the input, the noise level is also not enlarged even after introducing the dither ν [the variance (18) is the same as (10) when $E[\eta^2]$ is ignored].

B. Nonsubtractively Dithered System

Subtractively dithered system requires synchronous dither subtraction at the system output, which may not always be implementable in practical situations. For such reason, the nonsubtractively dither technique in the following is also of interest.

Nonsubtractively dithered system for current quantization is shown in Fig. 6(b). The current measurement error is expressed by

$$\begin{aligned} e_i &= i_m - i \\ &= Q\big(i_\eta + \nu\big) - i_\eta + \eta \\ &= q\big(i_\eta + \nu\big) + \nu + \eta. \end{aligned} \tag{19}$$

According to the Theorem in [20], a dither with the triangular probability density function (PDF)

$$p_\nu(\epsilon) = \begin{cases} \frac{1}{\Delta^2}(\epsilon + \Delta), & -\Delta < \epsilon \leq 0 \\ \frac{1}{\Delta^2}(\epsilon - +\Delta), & 0 < \epsilon \leq \Delta \\ 0, & \text{otherwise} \end{cases} \tag{20}$$

is the unique choice to render the mean and the variance of $q(i_\eta + \nu) + \nu$ independent of the input i_η with a minimal noise level. In this case, mean and variance of ν and e_i are expressed as:

$$E[\nu]$$

2) *Metering Error is Gaussian Noise:* In this case, the optimal design of ν by (27) is unavailable. Therefore, a suboptimal design of ν is considered. The following proposition is achieved:

Proposition 2: Suppose that the metering error η is Gaussian noise and has the Normal distribution

$$p_\eta(\epsilon) = \frac{1}{\sqrt{2\pi\sigma^2}}\exp\left\{-\frac{1}{2}\frac{\epsilon^2}{\sigma^2}\right\}, \qquad 0 < \sigma^2 < \frac{\Delta^2}{6} \tag{31}$$

where σ is the standard deviation. Then, w is Gaussian with

$$p_w(\epsilon) = \frac{1}{\sqrt{2\pi\frac{\Delta^2}{6}}}\exp\left\{-\frac{1}{2}\frac{\epsilon^2}{\frac{\Delta^2}{6}}\right\} \tag{32}$$

and has the same variance with the triangular dither (20) if p_ν satisfies

$$p_\nu(\epsilon) = \frac{1}{\sqrt{2\pi(\frac{\Delta^2}{6}-\sigma^2)}}\exp\left\{-\frac{1}{2}\frac{\epsilon^2}{\frac{\Delta^2}{6}-\sigma^2}\right\}. \tag{33}$$

Proposition 2 can be proved according to the Convolution Theorem. In the case of $\sigma^2 \geq \frac{\Delta^2}{6}$, η can be regarded to be large enough to whiten the quantization error, so that ν becomes unnecessary. Fig. 8(a) illustrates the sum of two variables from the view of probability perspective. By calculation, the cumulative probability of (32) located in $[-\Delta,\ \Delta]$ is about 98.6%.

The comparison between the Normal distribution and the triangular distribution with the same variance is shown in Fig. 8(b). For practical situation, a dither with Normal distribution has almost the same advantage to whiten the quantization error. A further insight into the discussion is given in [21].

Even the metering noise can be exploited to reduce the total noise level in nonsubtractively dithered system, the additional noise may result in no improvement in current control. In Section III-C, Kalman filter is designed to estimate the original current signals from the dithered signals.

C. Kalman Filter for Noise Suppression

A state-space realization of the model (5) is expressed as [17]

$\dot{\mathbf{x}}$

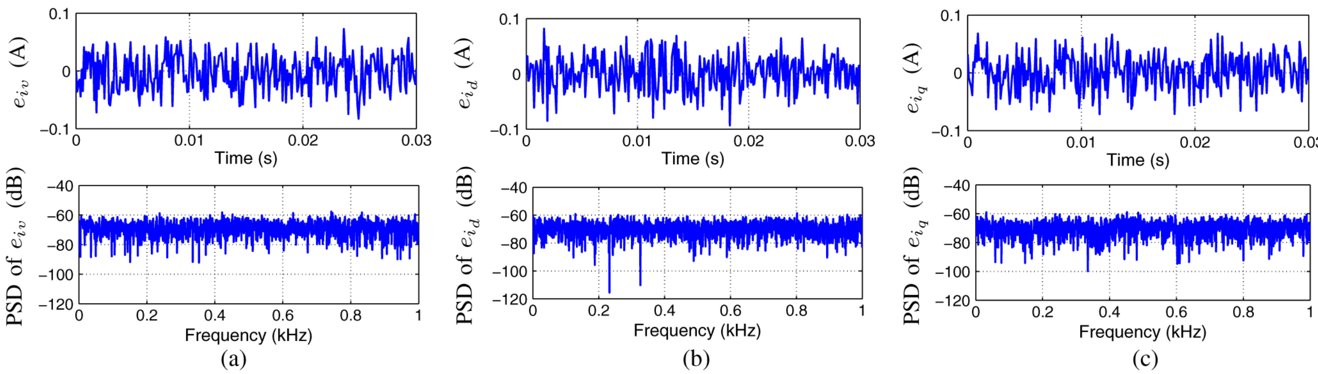

Fig. 12. Simulation results of Case 3 with subtractive dither and without Kalman filter. (a) Comparison of V-phase current. (b) Comparison of d-axis current. (c) Comparison of q-axis current.

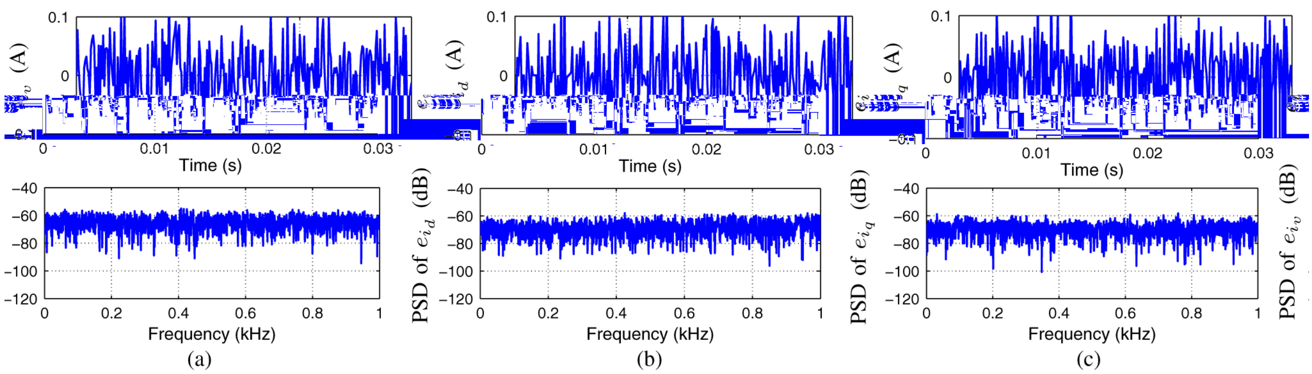

Fig. 13. Simulation results of Case 4 with nonsubtractive dither and without Kalman filter. (a) Comparison of V-phase current. (b) Comparison of d-axis current. (c) Comparison of q-axis current.

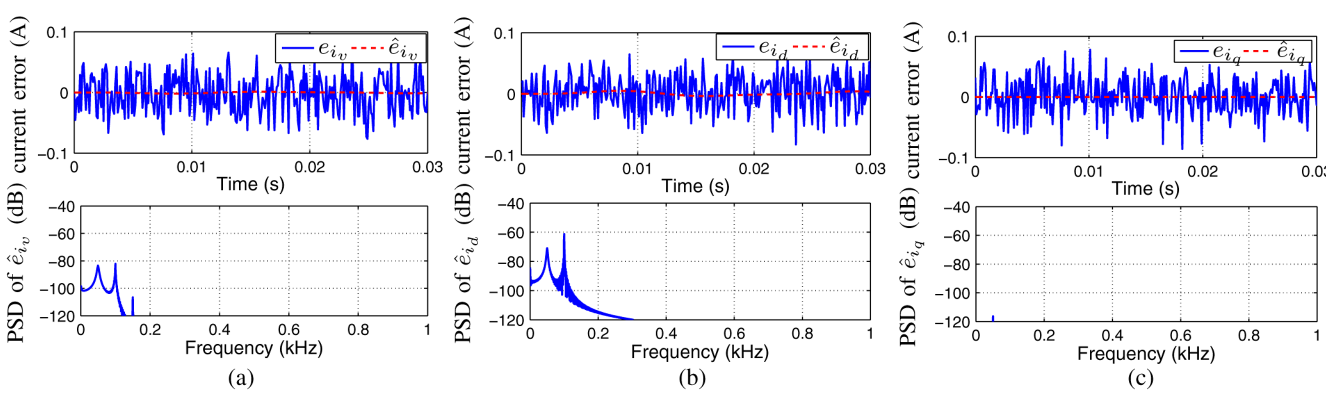

Fig. 14. Simulation results of Case 5 with subtractive dither as well as Kalman filter. (a) Comparison of V-phase current. (b) Comparison of d-axis current. (c) Comparison of q-axis current.

The results of dithering techniques combined with Kalman filter are shown in Figs. 14 and 15. The estimation errors $\hat{e}_{i_{v,d,q}}$ are far smaller than the measurement errors $e_{i_{v,d,q}}$, which means that the designed Kalman filter can estimate the current signals with high accuracy. Since Kalman filter also introduces phase delay, first-order resonances are also introduced. However, compared to the results of Case 2, the peak values of the resonances are relatively small. The comparison of the current errors evaluated via root mean square (RMS) and maximum value of PSD is shown in Table II.

The characteristics of the torque error of each case calculated by (15) is shown in Fig. 16. The torque ripples cannot be suppressed by low-pass filter. Moreover, the phase delay introduced by the filter worsens the torque ripples, as shown in Fig. 16(b). Fig. 16(c) and (d) show that the torque error can also be whitened by dithering techniques. In addition, the torque error

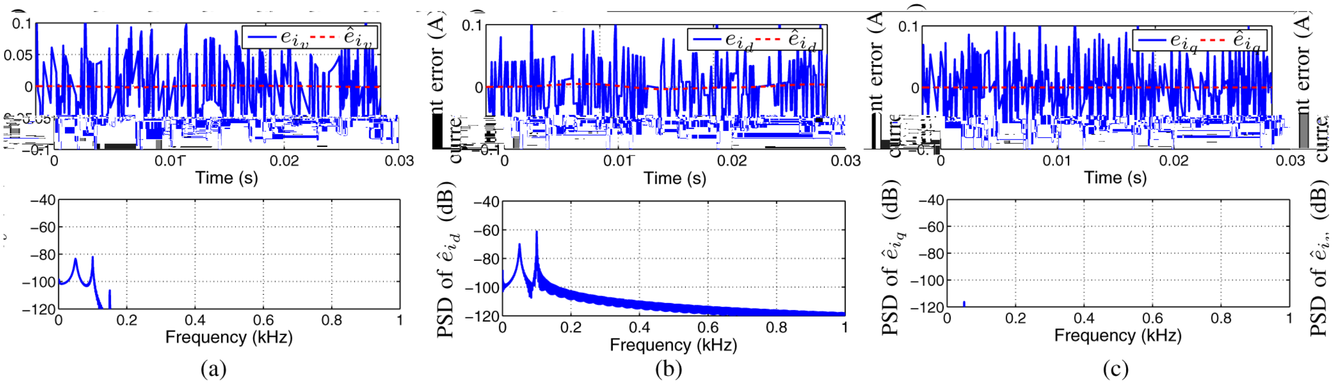

Fig. 15. Simulation results of Case 6 with nonsubtractive dither as well as Kalman filter. (a) Comparison of V-phase current. (b) Comparison of d-axis current. (c) Comparison of q-axis current.

TABLE II
COMPARISON OF THE CURRENT MEASUREMENT ERRORS (SIMULATION RESULTS)

Situations		Case 1	Case 2	Case 3	Case 4	Case 5	Case 6
$e_{i_v}/\bar{e}_{i_v}/\hat{e}_{i_v}$	RMS ($\times 10^{-2}$)	3.05	2.38	3.14	4.91	$2.20e^{-2}$	$2.42e^{-2}$
	PSD$_{\text{peak}}$ [dB]	-52.9	-41.4	-57.7	-55.0	-82.2	-81.6
$e_{i_d}/\bar{e}_{i_d}/\hat{e}_{i_d}$	RMS ($\times 10^{-2}$)	3.33	2.69	3.17	4.91	0.113	0.120
	PSD$_{\text{peak}}$ [dB]	-47.8	-51.9	-58.7	-58.2	-61.2	-61.0
$e_{i_q}/\bar{e}_{i_q}/\hat{e}_{i_q}$	RMS ($\times 10^{-2}$)	3.11	2.45	3.14	4.93	$2.45e^{-3}$	$2.96e^{-3}$
	PSD$_{\text{peak}}$ [dB]	-53.5	-42.6	-58.9	-57.9	-116.3	-116.1

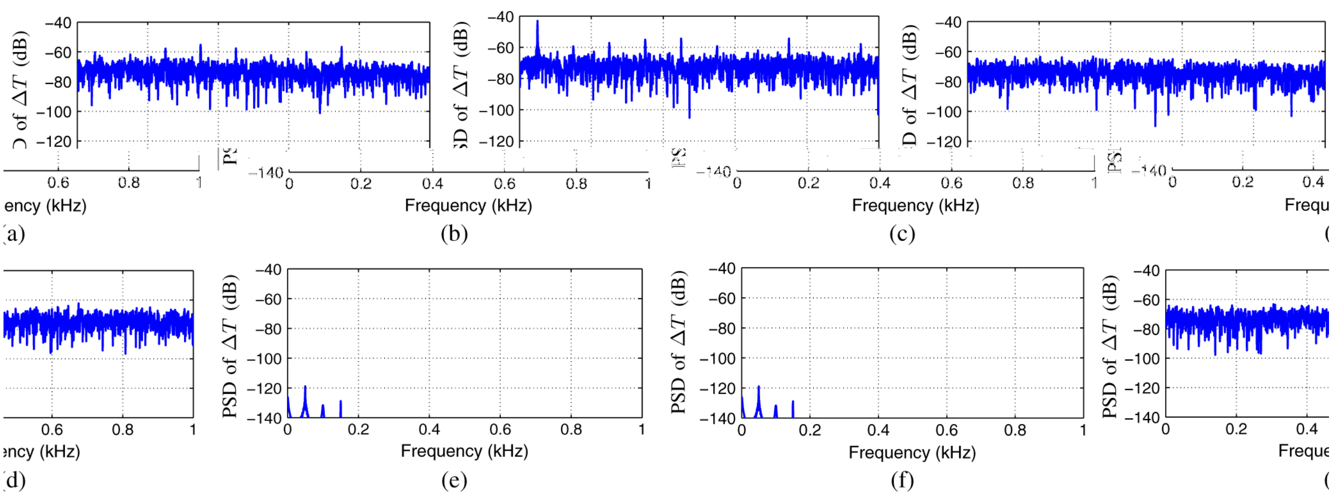

Fig. 16. Torque ripples generated by current measurement error: (a) Case 1; (b) Case 2; (c) Case 3; (d) Case 4; (e) Case 5; and (f) Case 6.

TABLE III
COMPARISON OF THE TORQUE ERRORS (SIMULATION RESULTS)

Situations	Case 1	Case 2	Case 3	Case 4	Case 5	Case 6
$\Delta T_{\text{PSD}_{\text{peak}}}$ [dB]	-55.27	-43.05	-63.30	-62.27	-118.9	-118.2

is significantly reduced by the methods combining dithering techniques and Kalman filter, as shown in Fig. 16(e) and (f). The comparison of the torque errors evaluated via the maximum value of PSD is shown in Table III.

From the results mentioned above, the combination of subtractively dithered technique and Kalman filter obtains the best performance. In the case where the synchronous dither subtraction is unavailable, the combination of nonsubtractively dithered technique and Kalman filter is desirable.

V. EXPERIMENTS

A. Description of the Experiment System

A high-precision stage used for experiments is shown in Fig. 17, and its parameters are shown in Table IV. The stage is a simplified model of the scanners in exposure systems used for the fabrication of integrated circuits. Two linear motors located at the both sides of the carriage are applied to drive the stage. An air guide system is introduced to reduce the friction between the

nt signals are
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he processor to imple-

loop is set as 1 kHz. A two-
exploited for position control
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control system is shown in Fig. 18.
ectiveness of the dithering techniques,
DCs is dropped to 10-bit by software, so
ent quantization step is $\Delta = 0.0977\text{A}$. For
ecise current probe (Tektronix) is used to
al current signal. Dither signals are generated in
ox–Muller method" [24] and then converted into
gnals via D/A converters with a resolution of 16-bit.

rst, the probability distribution of the metering noise is investigated. A current signal measured by current measurement system without executing the control system is used for analyzing the probability characteristic of the metering noise η. A sample of η in time domain is shown in Fig. 19(a) and the histogram plot is shown in Fig. 19(b). The metering noise can be approximated with Gaussian noise. By calculation, the variance of η is about

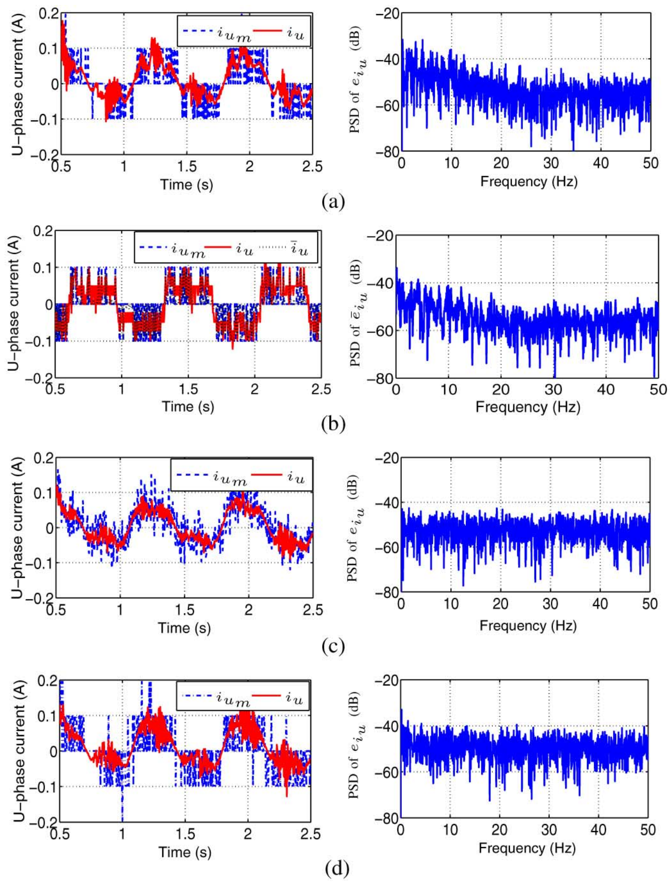

Fig. 21. Experimental results of Case 1-4. Graph (a) shows the results of Case 1 that the harmonics caused by quantization appear especially at low frequencies. Graph (b) shows the results of Case 2 that although the level of measurement noise can be somewhat reduced by low-pass filter, the harmonics cannot be suppressed. Graphs (c) and (d) show the results of Case 3 and Case 4, respectively. The results indicate that the harmonics caused by quantization noise are suppressed by the dithering techniques.

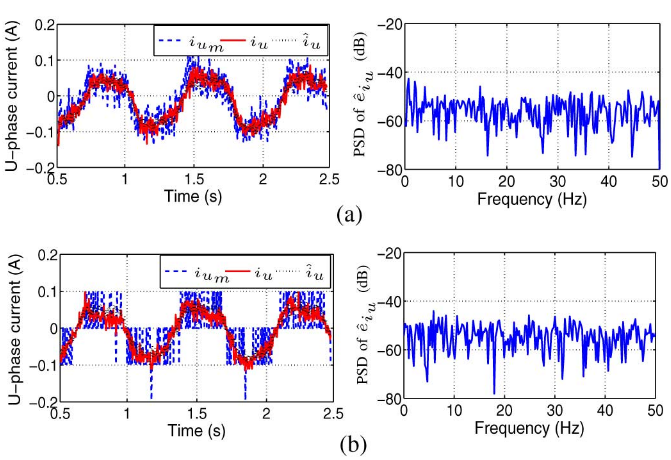

Fig. 22. Experimental results without applying Kalman filter. (a) Experimental results of Case 5. (b) Experimental results of Case 6.

with low-pass filter, dither techniques, and Kalman filter, respectively, is shown in Fig. 24. The maximum position tracking errors of all six cases are summarized in Table V. From the results, the combination of subtractively dithered technique and Kalman filter can obtain minimum position tracking error.

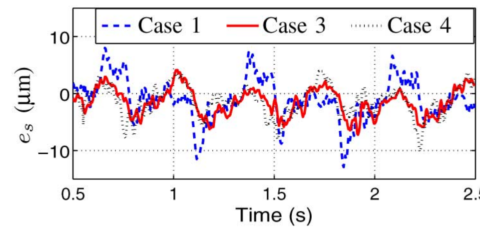

Fig. 23. Comparison of position tracking errors of Cases 1, 3, and 4, where no filter is applied.

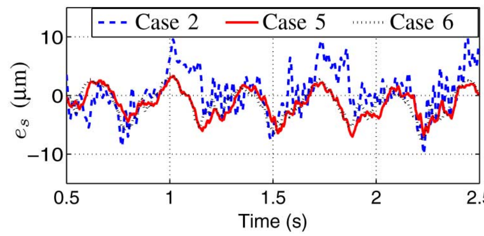

Fig. 24. Comparison of position tracking errors of Cases 2, 5, and 6, where filters are applied.

TABLE V
COMPARISON OF POSITION TRACKING ERROR (EXPERIMENTAL RESULTS)

Case	1	2	3	4	5	6
RMS of e_s $[\mu m]$	3.81	3.78	2.83	3.01	2.78	2.79
Max. of e_s $[\mu m]$	12.90	10.74	7.23	10.21	7.03	7.81

VI. CONCLUSION

In this paper, the combination of dithering techniques and Kalman filter for suppressing current quantization effects is proposed. In order to whiten the quantization error and reduce the total measurement noise level simultaneously, the probability characteristics of the metering noise is utilized to design the nonsubtractively dithered system. Furthermore, Kalman filter is applied to estimate the real current signals from the dithered measurement signals. Since the variance of the total measurement error is calculable, the covariance matrix for Kalman filter becomes explicit. The simulation and experimental results demonstrate the effectiveness of the proposed method.

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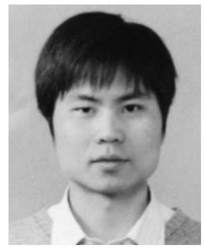

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